

Week 1

Exercise 2: Implementing the Factory Method Pattern

Code:

```
J Main.java J Factory.java 2, U X
J Factory.java > Language Support for Java(TM) by Red Hat > Factory
1  interface Shape {
2      void draw();
3  }
4
5  class Circle implements Shape {
6      public void draw() {
7          System.out.println(x: "Circle");
8      }
9  }
10
11 class Rectangle implements Shape {
12     public void draw() {
13         System.out.println(x: "Rectangle");
14     }
15 }
16
17 class ShapeFactory {
18
19     public Shape getShape(String type) {
20
21         if(type.equalsIgnoreCase(anotherString: "circle"))
22             return new Circle();
23
24         return new Rectangle();
25     }
26 }
27
28 public class Factory {
29     Run main | Debug main | Run | Debug
30     public static void main(String[] args) {
31
32         ShapeFactory factory = new ShapeFactory();
33
34         Shape s = factory.getShape(type: "circle");
35
36         s.draw();
37     }
38 }
```

Output:

```
PS D:\DN tasks>
PS D:\DN tasks> d:; cd 'd:\DN tasks'; & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessage'
User\workspaceStorage\17d3a9962cc7fc0c8ac76662ff4cf0a9\redhat.java\jdt_ws\DN tasks_111505f1\bin' 'Factory'
Circle
PS D:\DN tasks> 
```